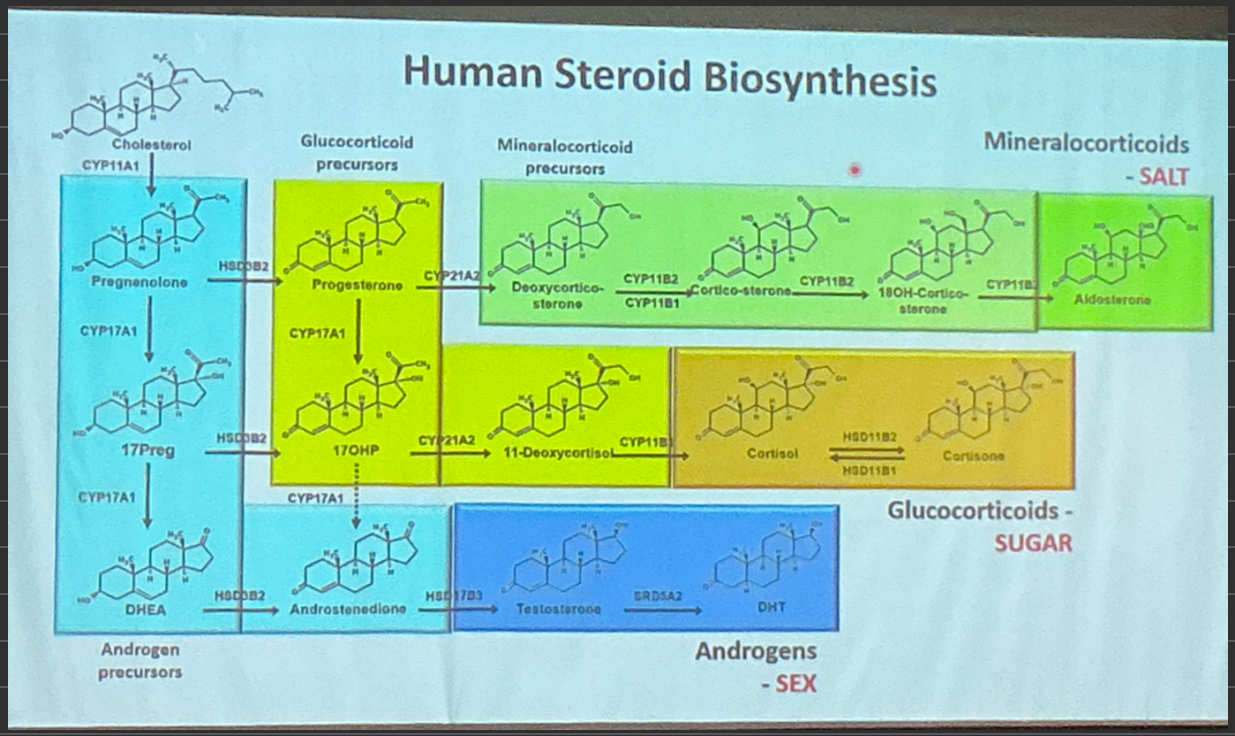
# ADRENAL DYSFUNCTION

# The Golden Rules of Endocrinology

1. All things are logical (if not, you might be on to some discovery)
2. Hormones are regulated by feedback
3. Think in diagnostic pairs
4. Hormone deficiency – stimulation test
5. Hormone excess – suppression test
6. Always think first, then examine clinically, then measure hormones and then do imaging
7. There are few but important exceptions to all rules, also to the golden ones

386 *


Human Steroid Biosynthesis
Cholesterol
CYP11A1
Glucocorticoid precursors
Mineralocorticoid precursors
Mineralocorticoids - SALT
Pregnenolone
HSD3B2
Progesterone
CYP21A2
Deoxycortico-sterone
CYP11B2
CYP11B1
Cortico-sterone
CYP11B2
18OH-Cortico-sterone
CYP11B2
Aldosterone
CYP17A1
17Preg
HSD3B2
17OHP
CYP21A2
11-Deoxycortisol
CYP11B1
Cortisol
HSD11B2
HSD11B1
Cortisone
Glucocorticoids - SUGAR
CYP17A1
CYP17A1
DHEA
HSD3B2
Androstenedione
HSD17B3
Testosterone
SRD5A2
DHT
Androgen precursors
Androgens - SEX

# Adrenal Cortex & Medulla

- Too little
  - Glucocorticoid deficiency
  - Mineralocorticoid deficiency
  - Adrenal androgen deficiency

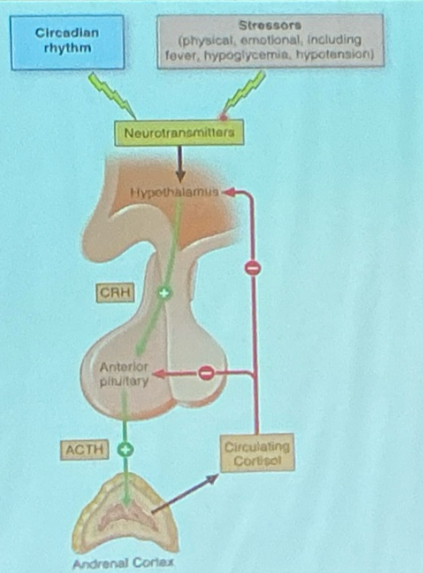
  } Adrenal Insufficiency

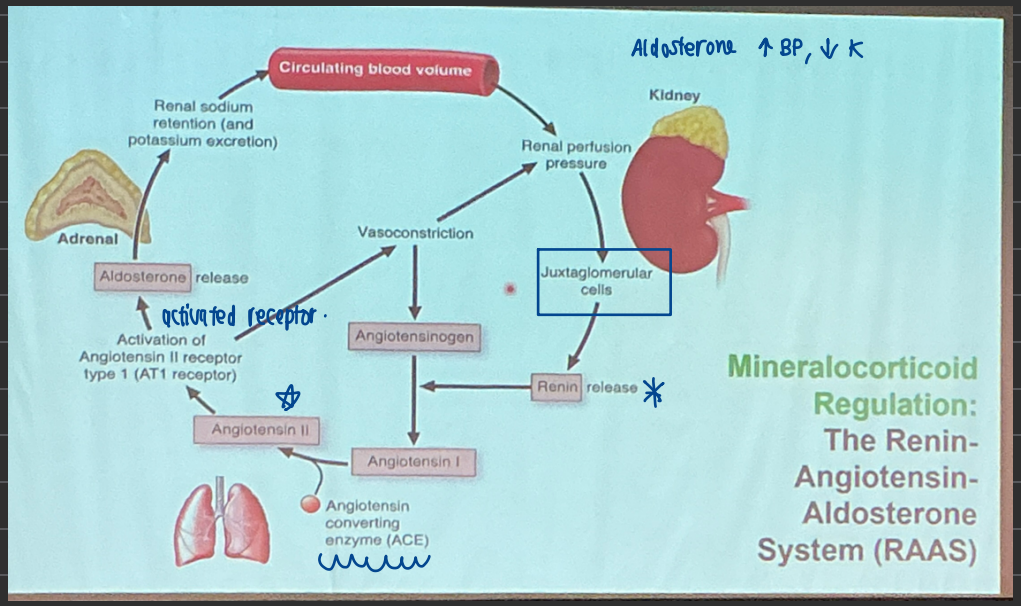
- Too much
  - Glucocorticoid excess } Cushing's syndrome
  - Mineralocorticoid excess } Conn's syndrome

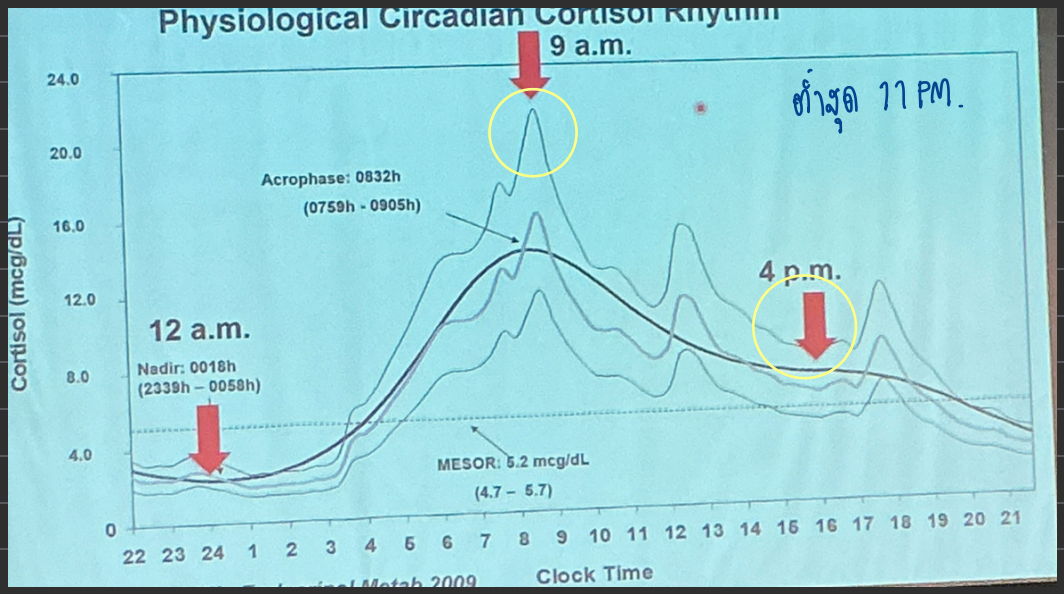
primary hyperaldosteronism.

- Adrenal Tumours
  - Adrenal incidentaloma, Adrenocortical Carcinoma
  - - Phaeochromocytoma (Catecholamine excess)
- Androgen excess
  - Congenital adrenal hyperplasia, Polycystic Ovary Syndrome

## Glucocorticoid Regulation: The Hypothalamic-Pituitary-Adrenal (HPA) Axis




Cortisol เกรงสุด ตอนเช้า *
(peak แล้วต่ำเวลม?)

# Too Little Cortisol – Adrenal Insufficiency

## SYMPTOMS ↓cortisol↓

- Weakness, fatigue (100%)
- Anorexia (100%)
- Gastrointestinal (86%)
  - nausea, vomiting, abdominal pain
- Salt craving (16%) อยากกินของเค็ม
- Postural dizziness (12%)
- Muscle/joint pains (10%)
- Loss of libido (women)

## SIGNS

- Weight loss (100%) ✓
- Hyperpigmentation (94%) ↑ ACTH ✓
- Hypotension (90%) [ACTH↑↑]* → MC1R
- Vitiligo (15%)
- Hyponatraemia (88%)
- Hyperkalaemia (64%)
- Hypercalcaemia (6%)
- Hypoglycaemia (5%)
- Uraemia (55%)
- Anaemia (40%)
- Loss of pubic hair (women)

Cortisol – Aldosterone – DHEA/Androgens

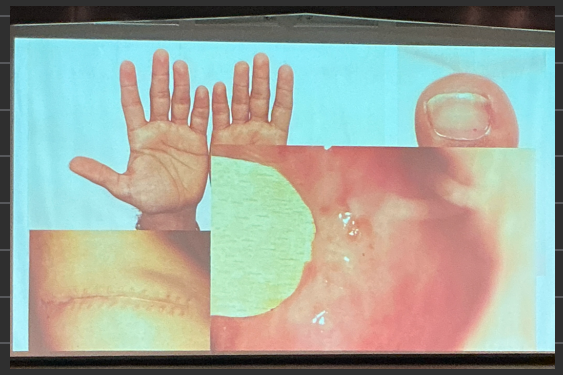
♂ ↓ testis ↓ Androgen

Addison's ← Autoimmune disease attack

พบ Vitiligo hypopigmentation

## Diagnosing Adrenal Insufficiency

- Colin, 18 years
- For last 2 years very tanned
- Last 6 months: fatigue, loss of appetite, dizziness, collapse, weight loss



1° AI เสีย Aldosterone ↓ $K^+$ (hypokalemia)

## Diagnosing Adrenal Insufficiency

- Colin, 18 years
- For last 2 years very tanned
- Last 6 months: fatigue, loss of appetite, dizziness, collapse, weight loss
- Lab results (July 2013):
  - Na 128 mmol/L (normal 134-146) ↓
  - K 4.7 mmol/L (normal 3.4-5.2) upper limit
  - Urea 12.1 mmol/L (3.2-7.6)
  - Creatinine 97 µmol/L (normal 6-126)

Cortisol investigation

↳ ส่งตอนเช้า.

## Diagnosing Adrenal Insufficiency

- Colin, 18 years
- For last 2 years very tanned
- Last 6 months: fatigue, loss of appetite, dizziness, collapse, weight loss
- Lab results (July 2013):
  - Cortisol (baseline ) 67 nmol/L (<150)
  - Cortisol 30' after ACTH stimulation 65 nmol/L (<450)
  - ACTH >2000 ng/L (normal 9-58)
  - Aldosterone <60 pmol/L (RR 111-860)
  - Renin >310 ng/L (RR 5.1-38.7)

ส่งตรวจ cortisol, Ald X ฟัง glomerulosa fasciculata

} DX Pair

ACTH สูง, cortisol ↓

} DX Pair

DHEA 0.L µmol ↓ เสีย 3 zone

Adrenal ที่ problems

ACTH stimulation test

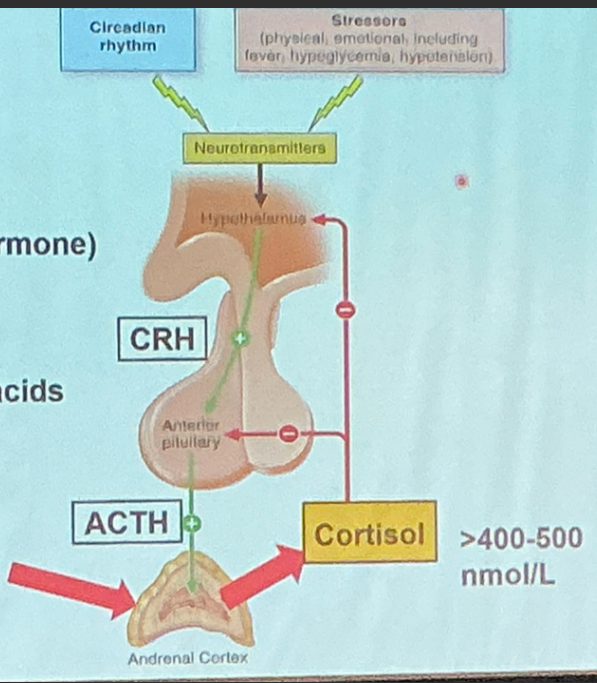
- response
- No response → 1° AI

## Diagnosing Adrenal Insufficiency: The Short Synacthen Test (SST)

$ACTH_{1-39}$ (adrenocorticotropic hormone)
= peptide with 39 amino acids

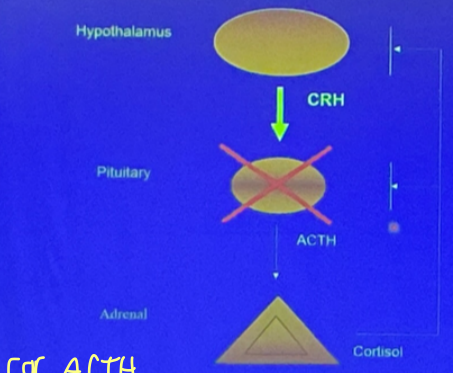
$ACTH_{1-24}$
= shorter peptide with 24 amino acids
= synacthen (Europe)
= cosyntropin (North America)

250ug $ACTH_{1-24}$ i.v. or i.m.



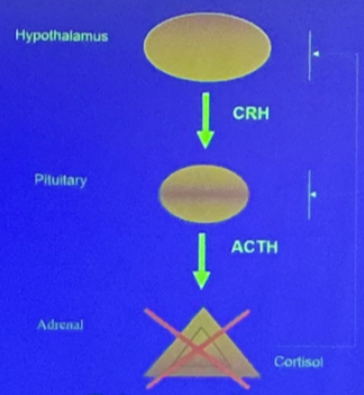

>400-500 nmol/L

## HPA Axis Regulation in Adrenal Insufficiency



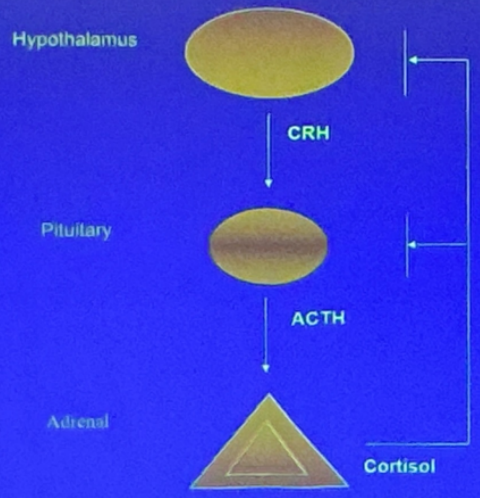

cor ACTH ↓ ↓

Secondary adrenal insufficiency

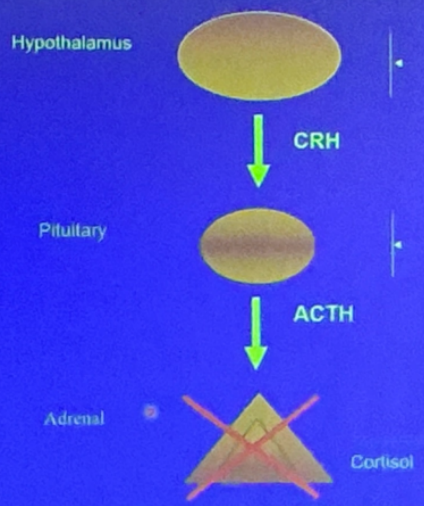



Primary adrenal insufficiency

## HPA Axis Regulation in Adrenal Insufficiency

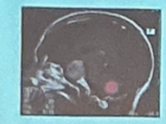



Physiological Situation

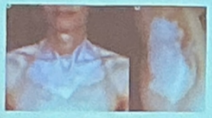



Primary adrenal insufficiency

## Adrenal Insufficiency – Who is affected?

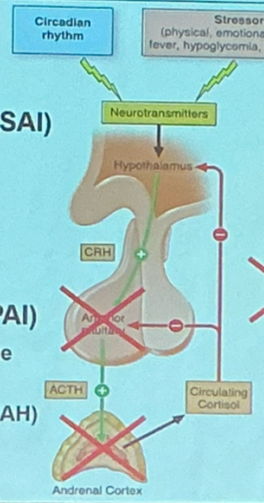


**SECONDARY AI (SAI)**
Hypopituitarism
3:10,000

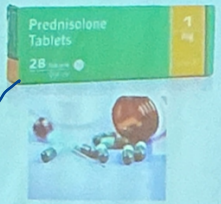


**PRIMARY AI (PAI)**
Addison's Disease
1:10,000

Congenital Adrenal Hyperplasia (CAH)
1:12,000




w/ Cushing's syndrome
SECONDARY AI
Iatrogenic AI




Cortisol ↓ HPA axis (Atrophy)

6-8 wk ⊖ Atrophy

# Differential Diagnosis of Adrenal Insufficiency

## Primary adrenal insufficiency (PAI)

- Most frequent cause: **autoimmune adrenalitis => measure adrenal (21-hydroxylase) auto-antibodies**
- If adrenal auto-antibodies negative and male patient: very long chain fatty acids (to exclude adrenoleuko-dystrophy/adrenomyeloneuropathy)
- If both negative: CT adrenals + exclude Tuberculosis
- Exclude **congenital adrenal hyperplasia**!

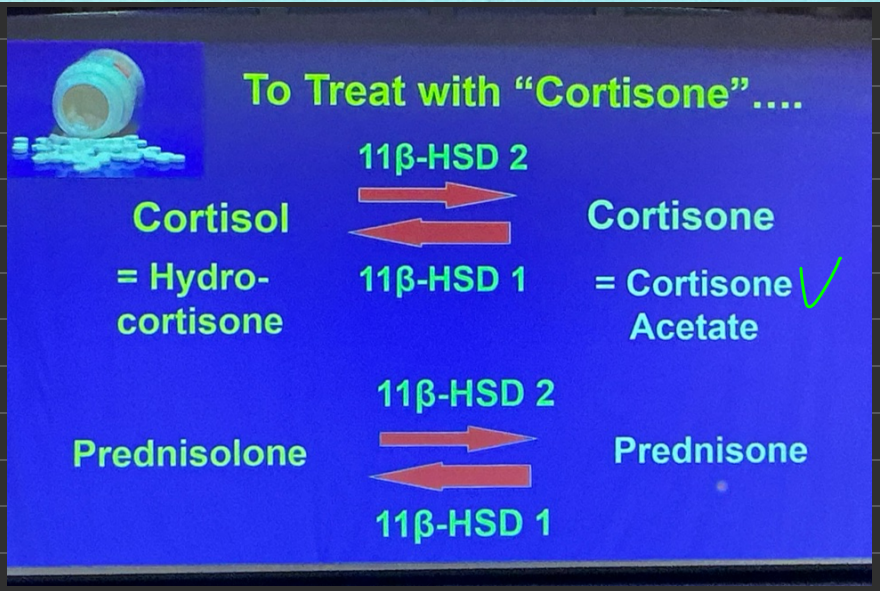
21 β- hydroxylase

## Secondary adrenal insufficiency (SAI)

exogenous, tumor กลุ่ม corticoids

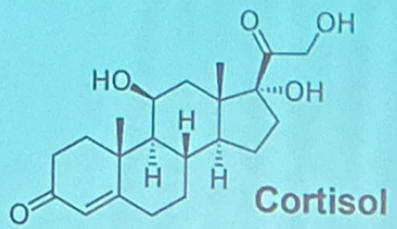
- Most frequent cause: **pituitary tumours** and their treatment
- ! RAAS intact - no aldosterone replacement needed
- ! Other hypothalamic-pituitary axes might be affected

2° AI cause pituitary tumors. ไม่ต้องดู Aldosterone เพราะ RAAS *

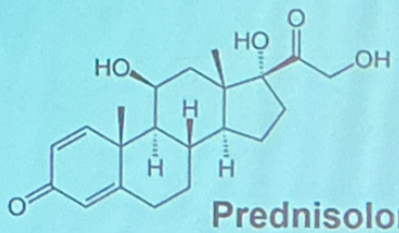


# Therapeutic corticosteroids

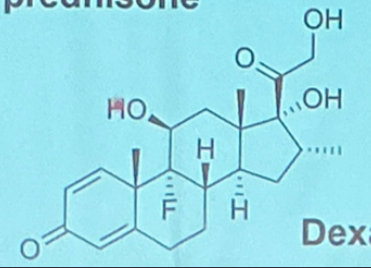
Equivalent anti-inflammatory effects
- 20 mg hydrocortisone → minero cor
- = 25 mg cortisone acetate min
- = 0.25 mg dexamethasone potency max
- = 4 mg prednisolone
- = 5 mg prednisone



Cortisol



Prednisolone



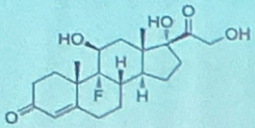
Dexamethasone

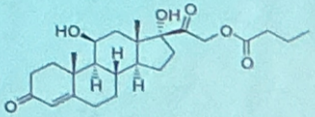
inactive dose สูงกว่า active เพื่อให้ equivalent

Dexamethasone High potency

# Mineralocorticoid Activity

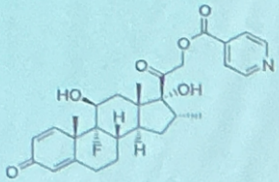
100 MineraloCorticoid Units (MCU)
= 100 µg Fludrocortisone
= 40 mg Hydrocortisone




=> If Daily Hydrocortisone Dose >50 mg, Fludrocortisone (FC) can be paused, otherwise patient with PAI take 100-250 µg FC/day

!! Dexamethasone has NO mineralocorticoid activity




V
Fludrocortisone

Dexamethasone
ไม่มี mineralocorticoid activity *

AI treatment *

## Treatment of adrenal insufficiency

**Long-term treatment**

- Replace glucocorticoid - Hydrocortisone 15mg AM, 10mg PM
- Replace mineralocorticoid – Fludrocortisone 100-200 µg/day
  (only in primary, not necessary in secondary adrenal failure)
- Adrenal androgens - DHEA (in women - low libido, low energy)

**Acute adrenal insufficiency**

**Rapid onset – Medical Emergency!!!! Fast reaction saves lives !!!** Treat shock – generous saline infusion, immediate injection of hydrocortisone 100mg, followed by continuous hydrocortisone infusion 200mg/24 h

ยาหลัก hydrocortisone – GC.
Fludrocortisone → MC

**Check out the CoMICs video on adrenal insufficiency:** These videos are created by a group of medical students and junior doctors and reviewed by experts.

Addison

**Episode 9:**
**Primary Adrenal Insufficiency**
https://www.youtube.com/watch?v=xPMO6EY2bAo

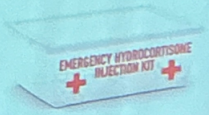
**Episode 75:**
**Secondary Adrenal Insufficiency**
https://www.youtube.com/watch?v=Ud2Jq7MiMZk

**Episode 76:**
**Glucocorticoid-Induced (Tertiary) Adrenal Insufficiency**
https://www.youtube.com/watch?v=0SEQfmfQnAE

To view more such CoMICs, please visit
Webpage: bit.ly/SimbaComics
Twitter: @SimbaComics
Instagram: @Simba.Comics

## Crisis Prevention in Adrenal Insufficiency Patients

- **Steroid card** and/or medic-alert bracelet to identify exogenous steroid dependence
- **Sick Day Rule 1 - Moderate stress**
  in case of fever, infection requiring antibiotics, minor surgery under local anaesthesia
  => double the daily glucocorticoid dose
- **Sick Day Rule 2 - Severe stress**
  for trauma, major surgery, persistent vomiting, colonoscopy, active labour
  – => 100mg hydrocortisone i.v., followed by continuous infusion of 200mg Hydrocortisone/24h
- Hydrocortisone Emergency Self-Injection Kit & Training




steroid card *

ผู้ป่วย AI

Blue steroid card

moderate stress → double dose

severe stress

100 mg hydrocortisone *

# Cushing's Syndrome

prolonged & inappropriate *
( fx )
เตี้ยลง จาก osteoporosis

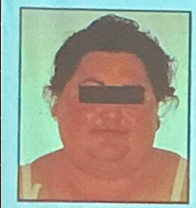
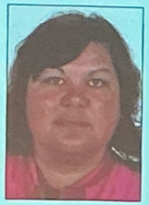
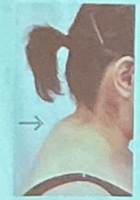
## Too Much Cortisol – Cushing's syndrome

### Definition of Cushing's syndrome

- "A constellation of symptoms & signs that reflect prolonged and inappropriately high exposure of tissues to glucocorticoids"
- Key words "prolonged", "inappropriately"

Consensus on diagnosis and management of Cushing's disease: a guidelineupdate.
Fleseriu M, Auchus R, Bancos I, Ben-Shlomo A, Bertherat J, Biermasz NR, Boguszewski CL, Bronstein MD, Buchfelder M, Carmichael JD, Casanueva FF, Castinetti F, Chanson P, Findling J, Gadelha M, Geer EB, Giustina A, Grossman A, Gurnell M, Ho K, Ioachimescu AG, Kaiser UB, Karavitaki N, Katznelson L, Kelly DF, Lacroix A, McCormack A, Melmed S, Molitch M, Mortini P, Newell-Price J, Nieman L, Pereira AM, Petersenn S, Pivonello R, Raff H, Reincke M, Salvatori R, Scaroni C, Shimon I, Stratakis CA, Swearingen B, Tabarin A, Takahashi Y, Theodoropoulou M, Tsagarakis S, Valassi E, Varlamov EV, Vila G, Wass J, Webb SM, Zatelli MC, Biller BMK.
**Lancet Diabetes Endocrinol. 2021** Dec;9(12):847-875. doi: 10.1016/S2213-8587(21)00235-7.

## Other (slightly less) specific features observed in Cushing's syndrome

Facial fullness and plethora

"Buffalo hump"

Harvey Cushing

**Osteoporosis** with vertebral fractures in the lumbar spine (loss in height)
**Decreased linear growth in children** = Weight gain + decreasing growth velocity

* * *

สังเกต
purplish striae * * osteoporosis *

Thin skin * easy bruising

proximal myopathy.
อ่อนแรง proximal * muscle tone & muscle power
Quadriceps.

## Clinical features most discriminating for Cushing's syndrome

**Purplish Stretchmarks Striae (>1cm)**

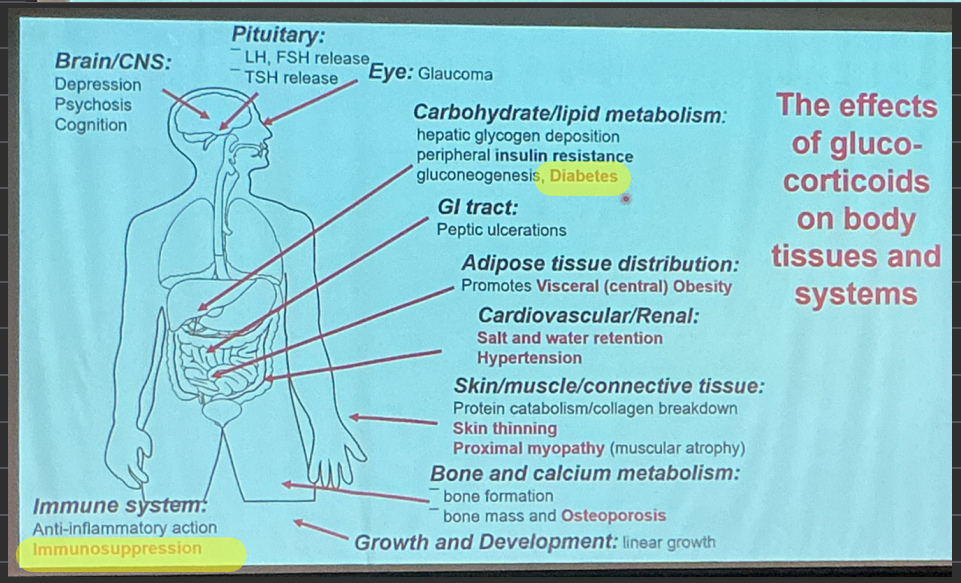
**Thin skin, easy bruising**

**Proximal myopathy**

Difficulties to get up from a chair unaided, difficulties climbing stairs

## Symptoms and Signs of Cushing's syndrome according to gender - 280 patients

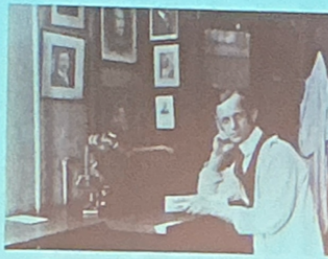
| Sign/Symptom | 233 Women (%) | 47Men (%) |
|---|---|---|
| Obesity | 86.2 | 85.1 |
| Moon Facies | 81.1 | 89.4 |
| Hypertension | 67.4 | 68.1 |
| Buffalo hump | 53.6 | 51.1 |
| Purple striae ✓ | 50.6 | 72.3 |
| Impaired glucose tolerance/diabetes | 45.4 | 42.8 |
| Proximal muscle atrophy ✓ | 45.5 | 63.8 |
| Psychiatric disturbance | 34.3 | 25.5 |
| Osteoporosis ✓ | 31.6 | 46.8 |
| Bruising / fragile skin ✓ | 32.2 | 21.3 |
| Acne | 27.6 | 19.1 |
| Hypokalemia | 21.8 | 32.6 |
| Nephrolithiasis | 6.4 | 21.3 |




## Too much glucocorticoid – Cushing's syndrome

Adrenal insufficiency *

- **Iatrogenic** – synthetic corticosteroids
  - ~1% of UK population taking chronic glucocorticoid therapy


Harvey Cushing (1869-1939)

- **ACTH-dependent Cushing syndrome**
  - **Pituitary dependent Cushing syndrome (= Cushing disease) (75%)**
  - **Ectopic Cushing due to ectopic ACTH secretion (15%)**
    - Small cell lung carcinoma
    - Carcinoids (occult – hidden – difficult to find)
- **ACTH-independent** = adrenal Cushing syndrome to to adrenocortical adenoma/carcinoma **(10%)**

ACTH
- Pituitary (Cushing's disease) *
- Ectopic cushing

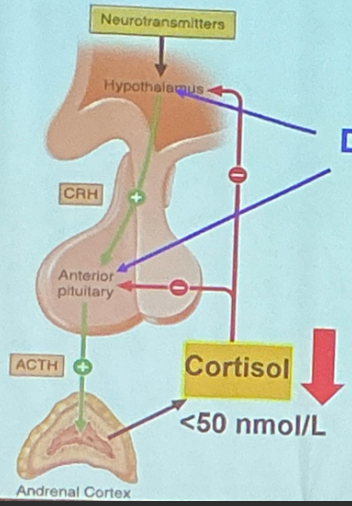
มีผลเรื่อง Salt & Water retention ด้วย.
เพราะ cortisol ↑ act like Aldosterone *

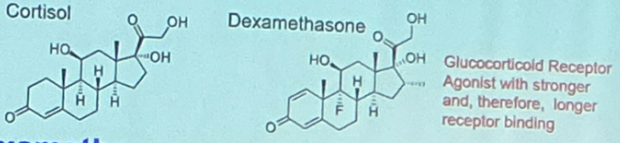
Cushing syndrome

ACTH- dependent
- pituitary
- ectopic

ACTH- independent adrenocortical adenoma

## Does my patient have cortisol excess?
## = Diagnosis of Cushing's syndrome




### Dexamethasone overnight suppression test

- 1mg dexamethasone tablet at 11 p.m.
- Blood for serum cortisol at 8-9 a.m.
- mimics cortisol effects by binding to glucocorticoid receptor but with longer half-life => ACTH is downregulated
- => endogenous cortisol suppressed
- In Cushing patients. cortisol secretion is autonomous and not responsive to feedback regulation => cortisol remains high

Low dose DST

Low-dose DST 1mg → Cushing's syndrome?

11 pm ⟶ ↓ cortisol ในเลือด 8-9 a.m.

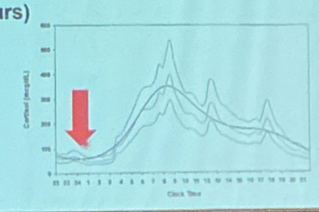
< 1.8 ug/dL ✓ suppres

## Does my patient have cortisol excess?
## = Diagnosis of Cushing syndrome

One of three highly sensitive tests

- **24 hour urinary free cortisol** (>130 ug/24h)
- **Dexamethasone suppression test**
  - Dexamethasone 1 mg overnight suppression test or
  - Low dose dexamethasone suppression test (***Liddle test***, 0.5mg 6-hourly for 48 hours)
- **"Midnight cortisol"** serum or saliva

เช้า: ตอนกลางคืน ยังสูง = Cushing ✓



Loss of cortisol circardian rhyme